

FORM 1
THE PATENTS ACT, 1970
(39 of 1970)

&
THE PATENTS RULES, 2003
APPLICATION FOR GRANT OF PATENT
[See sections 7,54 & 135 and rule 20(1)]

(FOR OFFICE USE ONLY)

Application No.:
Filing Date:
Amount of Fee Paid:
CBR No.:
Signature:

1. APPLICANT(S):

Sr.No.	Name	Nationality	Address	Country	State	Distict	City
1	BALAJI S	India	191-C, Ayyanar Appan Nagar, Palani Road, Oddanchatram, Dindigul, Tamil Nadu - 624619	India	Tamil Nadu		
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4	SABARISH R	India	29/17 kamatchiamman kovil street, south kumaranandhapuram, tiruppur, 641602.	India	Tamil Nadu	Tiruppur	Tiruppur

2. INVENTOR(S):

Sr.No.	Name	Nationality	Address	Country	State	Distict	City
1	BALAJI S	India	191-C, Ayyanar Appan Nagar, Palani Road, Oddanchatram, Dindigul, Tamil Nadu - 624619	India	Tamil Nadu	Dindigul	Oddanchatram
2	SELVAKUMAR M	India	Selva Kumar M, Assistant Professor, Department of Information Technology, Bannari Amman Institute of Technology,	India	Tamil Nadu	Erode	Sathyamangalam

			Sathyamangalam - 638401, Tamil Nadu, India.				
3	SELVA PRADEEP K	India	Anna nagar check post 1st Street,oddanchatram, Dindigul	India	Tamil Nadu	Dindigul	Oddanchatram
4	SABARISH R	India	29/17 kamatchiamman kovil street, south kumaranandhapuram, tiruppur, 641602.	India	Tamil Nadu	Tiruppur	Tiruppur

3. TITLE OF THE INVENTION: STUDENT ATTENDANCE TRACKING AND ANALYTICS SYSTEM

4. ADDRESS FOR CORRESPONDENCE OF APPLICANT / AUTHORISED PATENT AGENT IN INDIA:

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5. PRIORITY PARTICULARS OF THE APPLICATION(S) FILED IN CONVENTION COUNTRY:

Sr.No.	Country	Application Number	Filing Date	Name of the Applicant	Tilte of the Invention
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6. PARTICULARS FOR FILING PATENT COOPERATION TREATY (PCT) NATIONAL PHASE APPLICATION:

International Application Number	International Filing Date as Allotted by the Receiving Office
PCT//	

7. PARTICULARS FOR FILING DIVISIONAL APPLICATION

Original (first) Application Number	Date of Filing of Original (first) Application
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8. PARTICULARS FOR FILING PATENT OF ADDITION:

Main Application / Patent Number:	Date of Filing of Main Application
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9. DECLARATIONS:

(i) Declaration by the inventor(s)

I/We ,BALAJI S,SELVAKUMAR M,SELVA PRADEEP K,SABARISH R, is/are the true & first inventor(s) for this invention and declare that the applicant(s) herein is/are my/our assignee or legal representative.

(a) Date: 23.03.2026

(b) Signature(s) of the inventor(s):



(c) Name(s): BALAJI S,SELVAKUMAR M,SELVA PRADEEP K,SABARISH R

(ii) Declaration by the applicant(s) in the convention country

I/We, the applicant(s) in the convention country declare that the applicant(s) herein is/are my/our assignee or legal representative.

(a) Date: 23.03.2026

(b) Signature(s) :



(c) Name(s) of the singnatory: BALAJI S,SELVAKUMAR M,SELVA PRADEEP K,SABARISH R,BALAJI S,SELVAKUMAR M,SELVA PRADEEP K,SABARISH R,BALAJI S,SELVAKUMAR M,SELVA PRADEEP K,SABARISH R

(iii) Declaration by the applicant(s)

- The Complete specification relating to the invention is filed with this application.
- I am/We are, in possession of the above mentioned invention.
- There is no lawful ground of objection to the grant of the Patent to me/us.
- I am/We are, the assignee or legal representative to true first inventors.

10. FOLLOWING ARE THE ATTACHMENTS WITH THE APPLICATION:

Sr.	Document Description	FileName
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I/We hereby declare that to the best of my/our knowledge, information and belief the fact and matters stated hering are correct and I/We request that a patent may be granted to me/us for the said invention.

Dated this(Final Payment Date): 23.03.2026Signature: 

Name: BALAJI S

To The Controller of Patents

The Patent office at

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FORM 2

THE PATENT ACT, 1970
(39 of 1970)
&
THE PATENTS RULES, 2003

PROVISIONAL SPECIFICATION

(See section 10 and rule 13)

**1 . TITLE OF
INVENTION**

STUDENT ATTENDANCE
TRACKING AND ANALYTICS
SYSTEM

2 . APPLICANT(S)

Name	Nationality	Address
BALAJI S	India	191-C, Ayyanar Appan Nagar, Palani Road, Oddanchatram, Dindigul, Tamil Nadu - 624619
SELVAKUMAR M	India	Selva Kumar M, Assistant Professor, Department of Information Technology, Bannari Amman Institute of Technology, Sathyamangalam - 638401, Tamil Nadu, India.
SELVA PRADEEP K	India	Anna nagar check post 1st Street, oddanchatram, Dindigul
SABARISH R	India	29/17 kamatchiamman kovil street, south kumaranandhapuram, tiruppur, 641602.

3. PREAMBLE TO THE DESCRIPTION**PROVISIONAL**

The following specification describes the invention.

*The Complete Specification accompanying the
application treated as a provisional specification
under Section 9(3)*

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TITLE : STUDENT ATTENDANCE TRACKING AND ANALYTICS SYSTEM

Field of The Invention

[0001] This invention relates to the field of Information Technology and more particularly to a student attendance tracking and analytics system for educational institutions.

[0002] The invention provides an automated platform for recording, managing, and analysing student attendance through session-based tracking mechanisms. It integrates role-based access control, real-time notifications, and configurable academic policies to improve the efficiency and reliability of attendance management.

[0003] The system is useful in schools, colleges, and universities for monitoring attendance, ensuring compliance with academic regulations, reducing manual errors, and enabling data-driven decision-making. The invention enhances transparency, accountability, and operational efficiency in academic environments.

Background of the Invention

[0004] The present invention relates to the field of attendance management systems and more particularly to an automated and intelligent attendance tracking platform that integrates biometric authentication, session-based monitoring, and data analytics for educational institutions.

[0005] Another limitation of existing attendance systems is the absence of a structured and controlled grace attendance mechanism. Current solutions do not provide a reliable method to allow attendance within a predefined grace period along with a proper verification and approval workflow, leading to inconsistencies and misuse.

[0006] Additionally, many existing systems lack integrated analytical capabilities for administrators and departments. Institutions often face challenges in monitoring attendance trends, identifying irregular patterns, and making informed decisions due to fragmented or non-intelligent reporting mechanisms.

[0007] In conventional attendance systems, attendance is typically recorded manually or through basic digital solutions that rely on manual input or simple identification mechanisms such as RFID cards or login credentials. These methods are prone to several limitations, including proxy attendance, human error, lack of real-time validation, and inefficiencies in managing large volumes of student data.

[0008] Although certain modern systems incorporate biometric authentication technologies such as fingerprint or facial recognition, they often lack session-wise granularity and intelligent validation mechanisms. Furthermore, such systems provide limited flexibility in handling exceptional scenarios such as late attendance due to valid academic or institutional activities.

[0009] Further, existing platforms do not provide a centralized web-based interface that seamlessly integrates biometric data capture, attendance tracking, administrative control, and student interaction within a unified system, resulting in increased administrative overhead and reduced operational efficiency.

[0010] Therefore, there exists a need for an improved attendance management system that overcomes the aforementioned limitations by providing a secure, automated, and intelligent solution with enhanced accuracy, transparency, and efficiency.

Objectives of the Present Invention

[0011] The principal object of this invention is to introduce a grace attendance mechanism with an approval workflow that allows controlled and validated attendance updates.

[0012] To provide an automated student attendance tracking and analytics system that improves accuracy, efficiency, and transparency in attendance management.

[0013] To determine permissible leave limits required to maintain minimum attendance requirements.

[0014] To provide real-time notification mechanisms for informing users about attendance status and system updates.

[0015] To provide an analytics module for calculating attendance percentage, absentee records, and eligibility status based on predefined academic thresholds.

[0016] To provide a session-based attendance recording mechanism with configurable time constraints for multiple sessions including forenoon, afternoon, and hostel sessions.

[0017] To implement role-based access control for secure and structured access to system functionalities.

[0018] To provide administrative control for managing users, configuring working days, holidays, and session timings.

[0019] To provide a scalable and centralized system capable of supporting multiple academic groups and institutions.

Summary of the Present Invention

[0020] The present invention provides an automated student attendance tracking and analytics system comprising a web-based platform with integrated authentication, database management, and role-based access control.

[0021] The system enables session-based attendance recording with predefined time limits and supports multiple attendance sessions. An analytics module processes attendance data to generate insights such as attendance percentage, eligibility status, and permissible leave calculations.

[0022] A grace attendance mechanism is incorporated to allow users to submit attendance requests within a predefined grace period, wherein such requests are subject to an approval workflow before being updated in the system.

[0023] The system further includes real-time notification features, centralized dashboards for monitoring and analysis, and administrative modules for configuring academic parameters. The invention provides a reliable, secure, and scalable solution

that enhances attendance management through automation, analytics, and controlled validation mechanisms.

Brief Description of the Accompanying Drawings

[0024] Figure 1: Illustrates the flow diagram of the Grace Attendance System, depicting the process of attendance marking within the predefined time, submission of grace request within the allowable grace period, administrative review, and approval or rejection workflow leading to final attendance update.

[0025] Figure 1: Illustrates the overall working flow of the student attendance tracking and analytics system, including user authentication, session-based attendance marking, data validation, analytics processing, and dashboard visualization.

Detailed Description of the Invention

[0026] The present invention relates to an automated student attendance tracking and analytics system designed to manage attendance processes in educational institutions.

[0027] In one embodiment, the system comprises a web-based platform developed using a backend framework and a frontend interface, integrated with a database for storing attendance records and user information. The system supports multiple user roles including students, faculty, and administrators through role-based access control.

[0028] The system enables attendance recording based on predefined sessions such as forenoon, afternoon, and hostel sessions. Each session is associated with configurable time limits defined by the administrator. Attendance is validated based on time constraints and user authentication to prevent unauthorized access and proxy attendance.

[0029] In one embodiment, the system includes a grace attendance mechanism, wherein users who fail to mark attendance within the predefined time are allowed to submit a request within a limited grace period. These requests include valid reasons and are

stored for administrative review. Upon approval, the attendance is updated in the database.

[0030] In another embodiment, the system includes an analytics module that calculates attendance percentage, total working days, and absentee records. The system further determines permissible leave limits required to maintain a minimum attendance threshold.

[0031] In one embodiment, the system generates real-time notifications to users upon successful attendance marking and other system events. The system maintains logs of attendance records, requests, approvals, and modifications for auditing purposes.

[0032] In another embodiment, the system provides administrative functionalities including user management, configuration of working days and holidays, and customization of session timings. The system also supports multiple academic groups with independent configurations.

[0033] In one embodiment, the system includes a centralized dashboard for visualization and analysis of attendance data, enabling monitoring at individual and institutional levels.

[0034] The invention provides a scalable, efficient, and secure solution for attendance management, overcoming limitations of traditional systems through automation, intelligent analytics, and controlled validation mechanisms.

Examples

Example 1: Grace Attendance Handling in Academic Scenario

In an educational institution, if a student is unable to mark attendance within the predefined session time due to participation in a placement or event activity, the system allows the student to register biometric attendance within the defined grace period and submit a grace attendance request by providing a valid reason. The request is stored in the system and forwarded to the administrator for review. Upon verification of the

biometric record and approval of the request, the attendance is updated in the database and reflected in the student's attendance records. This ensures fairness and controlled validation of attendance updates.

Example 2: Attendance Analytics and Safe Leave Planning

A student accesses the system dashboard to monitor their attendance status. The system calculates the total working days, attended sessions, and attendance percentage. Based on a predefined threshold of 80%, the system determines the number of permissible leave days the student can take without affecting eligibility. This helps the student in planning leave effectively while maintaining the required attendance criteria.

Example 3: Administrative Configuration and Monitoring

An administrator configures working days, holidays, and session timings for different academic groups using the system. The system dynamically updates attendance calculations based on these configurations. The administrator can also monitor attendance records, review grace requests, and analyze reports through a centralized dashboard, ensuring efficient academic management.

We claim:

Claim 1 (Grace Attendance with Approval Workflow)

Our web portal enables a Grace Attendance System that allows students who are unable to register biometric attendance within the predefined time for forenoon or afternoon sessions, due to valid reasons such as placement or event participation, to mark attendance within a limited grace period of 30 minutes after the deadline. For example, if the normal attendance time is from 8:00 AM to 9:00 AM, the grace period extends from 9:00 AM to 9:30 AM. Attendance marked during this grace period is not directly updated in the main attendance records. Instead, students are required to submit a formal request through the portal specifying the reason. These requests are reviewed by the administrator, and only upon approval, the attendance is officially recorded in the attendance database, ensuring controlled, secure, and authenticated updates.

Claim 2 (Safe Leave Prediction – 80% Rule)

The system as claimed in claim 1, wherein the system is configured to calculate attendance percentage based on total working days and attended sessions and determine the number of permissible leave days a user can take while maintaining a minimum attendance threshold of 80%.

Claim 3 (Real-Time Notifications and Logs)

The system as claimed in claim 1, wherein the system sends real-time alerts to users upon successful attendance registration and maintains a detailed log of attendance entries, grace requests, approvals, and modifications for audit and tracking purposes.

Claim 4 (Dynamic Working Day Configuration)

The system as claimed in claim 1, wherein the system allows an administrator to define and modify working days and holidays, supports independent configuration for multiple academic groups or year levels, and updates attendance calculations dynamically based on the configured schedule.



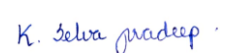
Selva Kumar M



Balaji S



Sabarish R



Selva Pradeep K

ABSTRACT

This project presents the design and development of a Student Attendance Tracking and Analytics System aimed at automating and enhancing traditional attendance management processes in educational institutions. The system is developed using Flask (Python) as the backend framework, along with Jinja, HTML, CSS, and JavaScript for the frontend, and integrates Firebase for secure Google-based authentication. The platform introduces a role-based access control mechanism, supporting three primary user roles: student, faculty, and administrator, each with specific permissions and functionalities.

The system enables efficient attendance recording through session-based tracking, including forenoon, afternoon, and hostel (night) sessions, with configurable time constraints defined by the administrator. It incorporates an intelligent analytics module that provides detailed insights into student attendance, including total working days, attendance percentage, absentee records, and eligibility status for end-semester examinations. Additionally, it calculates the permissible number of safe leaves required to maintain a minimum attendance threshold, thereby assisting students in academic planning.

A key feature of the system is the Grace Attendance Module, which allows students to request attendance consideration within a limited grace period in case of valid reasons such as academic or placement activities. This request undergoes an approval workflow managed by faculty or administrators, ensuring accountability and fairness. The system also includes a hostel dashboard for monitoring residential student attendance separately and a department analytics dashboard for filtering and analysing attendance data based on various parameters such as date and academic year.

The system integrates an email module that sends real-time email alerts upon attendance registration. Administrative functionalities include managing users, configuring working days and holidays for different academic years, and setting session timings. All data is stored in a structured database, ensuring scalability, data integrity, and efficient retrieval.

Keywords

Attendance Management System, Role-Based Access Control, Student Analytics, Biometric/Session-Based Attendance, Grace Attendance System, Attendance Data Analytics

GRACE ATTENDANCE SYSTEM

CONTROLLED AND AUTHENTICATED UPDATES

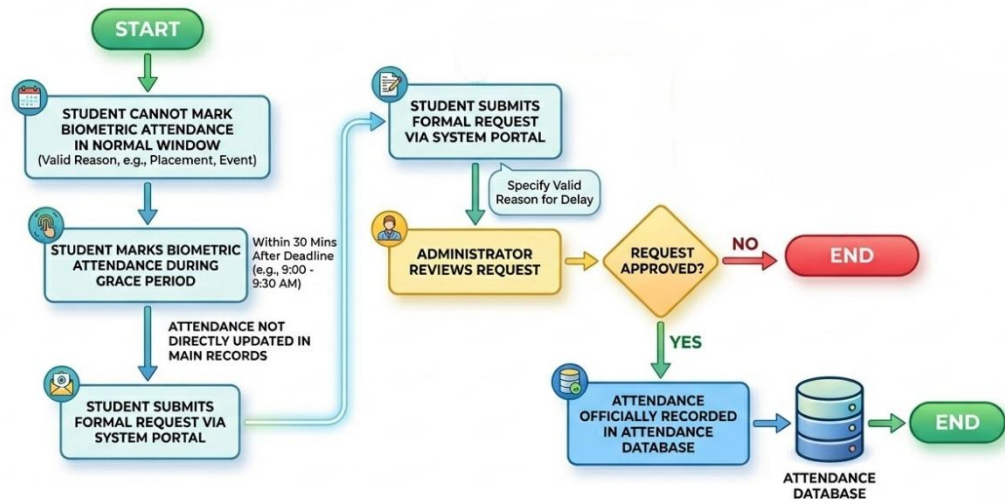


Figure 1 : Grace System

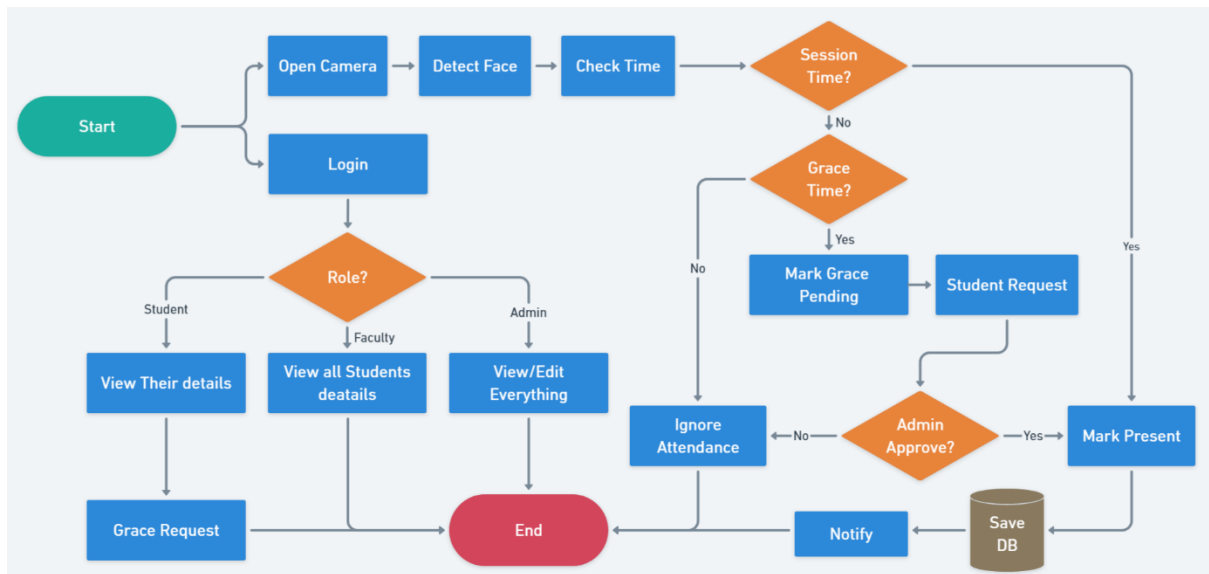


Figure 2 : Project Flow

Selva Kumar M

Selva Kumar M

Balaji S

Balaji S

Sabarish R

Sabarish R

K. Selva pradeep

Selva Pradeep K

FORM 5

THE PATENT ACT, 1970
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DECLARATION AS TO INVENTORSHIP

[See section 10(6) and rule 13(6)]

1. NAME OF APPLICANT(S) BALAJI S, SELVAKUMAR M, SELVA PRADEEP K, SABARISH R,

hereby declare that the true and first inventor(s) of the invention disclosed in the complete specification filed in pursuance of my/our application numbered **202641034597** dated **23/03/2026** is/are

2. INVENTOR(S)

Name	Country	Nationality	Address
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Dated this. **23/03/2026**Day of **2026**

Signature

Name of the signatory

3. DECLARATION TO BE GIVEN WHEN THE APPLICATION IN INDIA IS FILED BY THE APPLICANT(S) IN THE CONVENTION COUNTRY:--

We the applicant(s) in the convention country hereby declare that our right to apply for a patent in India is by way of assignment from the true and first inventor(s).

Dated this. **23/03/2026**.Day of **2026**

Signature

Name of the signatory

4. STATEMENT (to be signed by the additional inventor(s) not mentioned in the application form)

I/We assent to the invention referred to in the above declaration, being included in the complete specification filed in pursuance of the stated application.

Dated this(Final Payment Date):-----

Signature

Name of the signatory

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